

Quantification of pollutant re-introduction through ventilation openings into a building: A benchmark based on wind-tunnel experiments

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ABSTRACT

In occupational risk prevention, collective protection measures based on ventilation to reduce exposure to chemical hazards generally consist of containing a pollutant at its source, capturing it, and releasing it outdoors, after purification when it is applicable. However, unfiltered pollutants that are released can be re-introduced through ventilation air intakes or windows of the same building or neighbouring buildings, depending on wind conditions. Predicting the amount of pollutant re-introduced remains very challenging while essential to implement appropriate risk control measures. In this context, an experimental benchmark has been designed to quantify the dispersion and re-introduction of a pollutant in a reduced-scale building model. It accounts for the wind flow around the building, the building mechanical ventilation system, the emission and tracking of a tracer gas in an atmospheric boundary layer wind tunnel (ABLWT). In addition to measurements of tracer gas concentration in the ventilation exhaust duct for 13 wind directions (0, 15, 30, 45, 50, 60, 75, 90, 135, 180, 225, 270 and 315°) and 3 wind speeds (6.4, 9.0 and 11.1 m/s at the building model height), a mapping of the concentration inside and outside the building at the ventilation air intakes is provided for the worst-case scenario of wind conditions. This experimental data is crucial for validating, under controlled conditions, pollutant dispersion models for both outdoor and indoor environments in a computational fluid dynamics (CFD) framework, as well as for assessing ventilation network simulations using zonal or nodal approaches.

1. Introduction

Human activities, industrial processes, and products handled and stored inside buildings may emit compounds that are hazardous for health. To avoid the exposure of building occupants to chemicals, the emitted pollutants are captured, transported through ventilation ducts then released outdoors [1–5]. This practice raises concerns about its impact on the outdoor environment: this generates local outdoor pollution, called “building exhalation” [6,7], which is supposed to rapidly dissipate in the atmosphere. However, the efficiency of this dissipation is closely related to weather conditions and terrain topography. For example, in urban areas like valley cities, stable weather conditions caused by terrain barrier and temperature inversion will result in very unfavourable diffusion conditions [8], as well as in the countryside where stable atmospheric conditions and local sources may result in severe air pollution events [9]. Another consequence of

pollutant release by ventilation exhaust is that these sources can degrade the quality of the air used for ventilation of buildings located in the surroundings. As the dilution phenomenon is not immediate, a pollutant plume is established in the vicinity of the exhaust stack, implying a possible contamination of ventilation air intakes [10–15]. The pollutant can either be introduced into another building, which leads to pollutant transfer, or be introduced into the building of its origin, which is called pollutant re-introduction, re-ingestion or re-entry. This situation poses a risk of exposure to chemicals that was initially prevented by implementing a local exhaust ventilation system. It is critical when a building is dedicated to various kinds of activities, which is frequent on campuses combining laboratories or industrial facilities with meeting rooms and office spaces. The risk assessment applied to such workplaces might incorrectly conclude that there is no chemical exposure to meeting rooms and office spaces, based on the assumption that the ventilation system supplies only fresh outdoor air. Hazard identification and

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exposure assessment being the first steps of a risk management process, their accuracy is crucial for implementing adequate measures of risk reduction [16]. The pollutant transfer has been studied in various configurations [17–21,15,22], but the pollutant re-introduction through mechanical ventilation is less reported in the literature [11,13,14]. Studies on closely related topics instead focus on the re-introduction of contaminants through open windows [23,24], the dilution of pollutants from exhaust stacks [25], or the relative positioning of outdoor air intakes and outlets [26].

In order to mitigate the risk of pollutant re-introduction, the first way is to suppress or reduce the pollutant source. This can be done by filtering the mixture before exhaust, which can be supplemented with the filtration of air intakes when using air handling units [3,5]. This performs well for particulate matter, although increasing installation, working, upkeeping and waste management requirements. The treatment of gaseous pollutants is much more complex and demanding than particulate matter [27], hence it is not systematically implemented in buildings. The gaseous pollutants are thus simply released into the outdoor atmosphere.

Another way to prevent the risk of pollutant re-introduction is to release it sufficiently far from the building of its origin, together with a minimal exhaust velocity requirement to avoid recirculation and interaction between the pollutant plume and the building. Stack height and exhaust velocity recommendations have been established in this context, based on empirical models [28,29]. The relevance of these models has been deeply studied and refined, but they remain inappropriate for direct application to complex cases [30–33].

The Computational Fluid Dynamics (CFD) technique can significantly help to investigate specific scenarios under well-controlled conditions while accounting for the real geometry of the buildings and their environment, perform parametric studies and, when validated, provide practical recommendations for existing or future buildings [30,34–37]. However, while the Reynolds-averaged Navier-Stokes (RANS) approach - most used but less accurate - can be based on extensive best practice guideline documents [38–41], for the large eddy simulations (LES) approach - computational demanding but more accurate - best practices are still largely missing, which is considered as serious limitation for high-quality LES in wind engineering [42]. To validate these models and use them in real-life situations, it is essential to test them first on controlled cases. Full-scale measurements might not be appropriate for validation because it is impossible to control some boundary conditions (e.g. wind conditions, temperature, humidity).

To overcome this issue, very few wind tunnel studies aimed at treating the interactions between external wind conditions and internal mechanical ventilation system have been performed, for instance in the context of nuclear safety [43–45] and asbestos abatement from building interior [46]. However, there is no wind tunnel experiment quantifying the effects of wind on the possible re-introduction of pollutants through the mechanical ventilation system. The lack of such experimental studies may be attributed to the required multi-physics approach, coupling and scaling the atmospheric boundary layer (ABL) wind and the internal mechanical ventilation system, together with gas tracing techniques.

In this context, we propose a reduced-scale building model enabling a more comprehensive characterization of pollutant dispersion, starting from its discharge into the atmosphere, followed by its wind-driven transport, its re-introduction into the building through ventilation air intakes, and finally its extraction by the mechanical ventilation system. Considering the building ventilation in the experiment is crucial because wind pressure on the ventilation network boundaries affects airflow rates and consequently pollutant fluxes in both indoor and outdoor environments [46,44]. To maintain controlled experimental conditions, the selected case study represents a simplified geometry of an isolated building equipped with a single-flow mechanical ventilation system. This ventilation system, commonly used in modern buildings, consists of passive air inlets generally placed above the building windows,

combined with mechanical exhaust through a fan. The building model also includes a calibrated air infiltration rate to account for the air exchange through leaks. This reduced-scale model has already been studied from an aerodynamic and airflow perspective, with comprehensive characterization of ventilation and external wind pressure coefficients on facades in an atmospheric boundary layer wind tunnel (ABLWT) [47,46]. Regarding the representation of the pollutant, a tracer gas was emitted continuously from the exhaust stack of the building model, with a density equivalent to that of the ambient air to neglect buoyancy effects as a first step. This assumption is supported by the conclusions of Moon *et al.* [14], stating that the pollutant re-entry was favoured by weak winds and ambient temperature close to the release temperature for the case they investigated. The extracted tracer gas concentration from the building's interior through the ventilation exhaust duct was measured for 13 wind directions and 3 reference wind speeds to study the influence of wind conditions on the proportion of reintroduced gas. For two selected configurations, a mapping of tracer gas concentration inside the building was also carried out to characterize its distribution within the internal volume of the model. Finally, localized measurements of tracer gas concentration at each air intake were performed under the wind conditions leading to the highest re-introduction proportion.

The demonstration of the ability to combine neutral atmospheric wind conditions, mechanical ventilation and pollutant dispersion into one experimental setup represents a first step towards analysing realistic cases in the near future for risk quantification and pollutant control. Such an experimental benchmark is indeed crucial to validate CFD simulations of pollutant dispersion outside and inside buildings, as well as nodal or zonal simulations of pollutant transport through ventilation systems. This paper is organized as follows: the reduced-scale building model including its ventilation system is first described, followed by wind characteristics in the wind tunnel and the representation of the pollutant by a tracer gas. The measured concentrations are then presented and discussed, starting with obtained pollutant concentrations in the ventilation exhaust duct, followed by the internal and external concentration distributions.

2. Experimental setup and methodology

2.1. Building model and ventilation system

An isolated cubical building with dimensions $18 \times 18 \times 18 \text{ m}^3$ is considered for the study. The upper half of the building contains a mechanically ventilated volume measuring $18 \times 18 \times 9 \text{ m}^3$. The ventilation system is designed to maintain an internal depressurization of -20 Pa and an air change rate of 6 h^{-1} . The ventilation system design, detailed in Jayakumari *et al.* [48], has been adapted here to represent a generalized ventilation setup incorporating an exhaust stack for pollutant emission. It corresponds to the typical ventilation of a large laboratory when one fume hood is turned on. The siting rationale for the exhaust stack was to follow a common practical approach, which consists in positioning exhaust outlets far from roof edges. In addition, we deliberately selected an asymmetric configuration, which is also realistic, as exhaust locations are rarely centred. For these reasons, the exhaust stack was placed halfway between a roof edge and the roof centre. The ventilation system consists of forty air inlets, an exhaust fan, and a roof-mounted stack. The specifications of these components are provided in Table 1. To enable testing various experimental configurations, geometric symmetry with respect to the approaching wind direction is avoided by positioning the stack off-centre on the roof, and arranging the ventilation inlets non-uniformly across the building facades. Consequently, each tested wind direction corresponds to a distinct configuration. In addition to the primary ventilation components, to account for the building's airtightness, sixteen leakage openings, distributed evenly across the four facades, are included.

To conduct wind tunnel experiments, the scaling methodology pro-

Table 1
Ventilation characteristics in full scale.

Parameter	Value
Internal ventilated volume	2916 m ³
Depressurization	-20 Pa
Air change rate	6 h ⁻¹
Extracted flow rate	18000 m ³ /h
Air inlets	N = 40, S = 0.03 m ² , L = 1 m
Stack	S = 0.045 m ² , L = 1 m
Stack location on building roof	6 m from edges on a roof corner
Stack flow rate	1221 m ³ /h

N: number of components, S: opening area, L: length of the component

posed by Le Roux *et al.* [44] is adopted for both the building and its ventilation components. Further details on applying this approach to the building with ventilation system under consideration can be found in the publications of Jayakumari *et al.* [46,48] and briefly summarized here. In this methodology, a mechanically ventilated building is represented as a network of interconnected nodes and branches. Nodes correspond to spaces such as rooms, junctions, or the external environment (serving as boundary conditions), where variables like pressure, temperature, and contaminant concentration are assumed to be uniform. Branches denote the ventilation system components, including air inlets, fans, and leaks. By applying the principle of mass balance at the nodes and the generalized Bernoulli equation along the branches, two essential dimensionless parameters are obtained, expressed as follows:

$$N_1 = \frac{R_{ref} S_{ref}^n}{(\rho_{ref} U_{ref})^{2-n}} \quad (2.1)$$

$$N_2 = \frac{P_{atm} S_{ref} L_{ref}}{\rho_{ref} V_{ref} U_{ref}^2} \quad (2.2)$$

where R_{ref} is the reference value for airflow resistance (kg²⁻ⁿ/m⁴/s²⁻ⁿ), S_{ref} for branch section area (m²), U_{ref} for velocity (m/s), ρ_{ref} for air density (kg/m³), L_{ref} for branch length (m), V_{ref} for the volume of the node (m³) and P_{atm} is the atmospheric pressure (Pa). The exponent “ n ” depends on the branch type and is equal to 2 for ducts (*i.e.*, air inlets), 1 for fans equipped with HEPA filter and between 1 and 2 for leaks. Ensuring that the two dimensionless parameters are equal for both the full-scale and the reduced-scale models guarantees geometric and kinematic similarity and also establishes the relationship between different scaling factors. Applying this methodology while considering the practical constraints of modelling ventilation components the following scaling factors are determined for the study: geometric scale (building), 1:40; velocity scale, $\sqrt{2}$; pressure scale, 2; volumetric flow rate scale, 1:180; time scale, 1:178; frequency scale, 178; ventilation-opening length scale, 1:126; and ventilation-opening area scale, 1:225.

The scaling factors are applied consistently to both the building and

its ventilation components. A schematic of the reduced-scale model is shown in Figure 1a. The building's reduced-scale dimensions are $0.45 \times 0.45 \times 0.225$ m³, resulting in a blockage ratio between 3.4 % and 4.8 % depending on the incidence angle. In the setup, the air inlets are modelled as circular openings with a diameter of 0.012 m, each fitted with a perforated mesh to accurately reproduce the aerodynamic resistance at reduced scale. Leakage is represented by 16 holes, four on each façade, each with a diameter of 0.003 m. The exhaust system comprises a mechanical ventilation fan located at the bottom of the ventilated volume, extracting air at a rate of 100 m³/h. This fan is connected to the building volume via a 1.7 m long pipe of diameter 0.1 m, as illustrated in Fig. 1b. The Reynolds number based on the duct diameter is 23,600. The stack is positioned on the roof, in a corner, with its centre at 0.15 m from the edges of the building. The stack has a length of 0.025 m and an inner diameter of 0.006 m in the reduced scale model. The stack is connected to the tracer gas supply from beneath the building model. The building model with its ventilation components is mounted at the centre of a 2.6 m diameter turntable, enabling tests across multiple wind directions. As mentioned previously, although the building has a cubical geometry, the ventilation components are asymmetrically arranged, making each wind direction unique. A geometry STEP file of the building model is available in the dataset provided by Gillmeier and Jayakumari [49].

The effectiveness of the scaling methodology is analysed by measuring internal pressure of the reduced-scale building model with the active mechanical ventilation system. A Scanivalve MPS4264 pressure scanner is used to measure pressure at the centre of each façade. Results showed an average depressurization value of -43.5 Pa, confirming the validity of the scaling methodology.

2.2. Wind-tunnel set-up and ABL characterization

The wind tunnel (WT) experiments are conducted in the atmospheric boundary layer wind tunnel (ABLWT) facility at Eindhoven University of Technology (TU/e), the Netherlands. The WT has a cross-sectional area of 3 m (width) $\times$ 2 m (height). Fig. 2 shows the down-scaled model building equipped with the ventilation system inside the tunnel. The incoming ABL flow, representative of wind flow over a moderately rough terrain [50] at a geometric scale of 1:40, is generated using floor roughness elements and vortex generators. Three spires, based on the design proposed by Irwin [51], with a base width of 0.5 m, a top width of 0.04 m, and a base extension are used as vortex generators. The roughness elements consist of numerous ‘L-shaped’ metallic elements of size 0.06×0.06 m² and 0.04×0.04 m² placed along a fetch length of 12.9 m and further extending onto the turntable and downstream of the building model as shown in Fig. 3a-b.

The approach flow characteristics of the ABL, namely the velocity components u , v , w (in the x , y , and z directions, as shown in Fig. 3a – b), are measured at the centre of the turntable (without building model present) at reduced-scale heights ranging from 0.01 to 0.795 m.

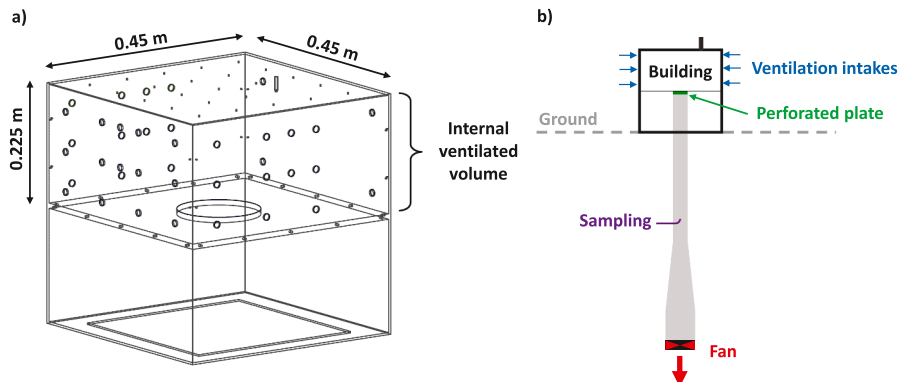


Fig. 1. a) 3D representation of the building model in reduced scale and b) schematic diagram of the reduced-scale ventilation system.



Fig. 2. Image of the building model with ventilation system installed in the test section of the ABLWT.

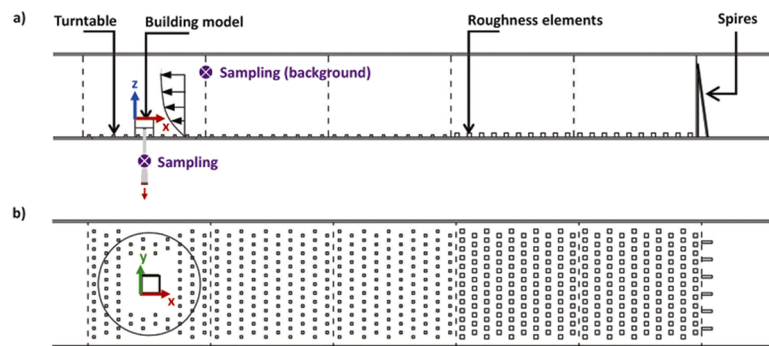


Fig. 3. a) Schematic side view of the test section, b) Schematic top view of the test section.

Instantaneous values of u , v , and w are recorded using a Series 100 Cobra probe at 1000 Hz for 120 s. Fig. 4 presents the vertical profiles of the streamwise mean wind velocity (U), turbulence intensity (I_u), and turbulence length scale (L_u) together with error bars. The logarithmic law fit to the mean velocity profile is shown in Fig. 4a. From the fit, the aerodynamic roughness length is estimated as 0.013 m and the friction velocity (u^*) as 0.4 m/s, corresponding to a *moderately rough* terrain category according to the VDI guideline [50]. Fig. 4b shows the turbulence intensity profile together with the VDI classification, where the measured I_u values also fall within the *moderately rough* terrain range. The streamwise turbulence length scales (L_u) expressed at full scale are plotted in Fig. 4c. The wind tunnel cross-section is 2 m (height) $\times$ 3 m (width) and the building model height is $H = 0.45$ m. It should be noted that in wind tunnels with limited cross-sectional dimensions, such as the present facility, it is inherently difficult to reproduce the largest turbulent structures and the turbulent length scale at a geometric scale of 1:40 due to blockage from the tunnel walls. This limitation has been discussed in detail in Jayakumari *et al.* [47]. Nevertheless, the chosen geometric scaling is inevitable, as the physical modelling of the ventilation components requires relatively large scales for practical implementation.

2.3. Pollutant release and measurements

Pollutant concentrations are measured using a two-channel HFR 400 Fast Flame Ionization Detector (FFID) from Cambustion Ltd. Images of

the FFID sensor and the capillaries at several measurement locations are shown in Fig. 5. Tracer gas concentrations at the measurement locations are detected by one channel of the FFID while the background concentration in the approach flow is simultaneously measured by the other channel, the location of which is shown in the schematic diagram of Fig. 3a. The gain of each FFID channel was set to 500, which is the highest sensitivity level. The HFR 400 FFID system measures the hydrocarbon concentration of a tracer gas. The gas enters a hydrogen flame where hydrocarbons are ionised, and the resulting ions are collected on an electrode under a high DC voltage. The generated current is proportional to the hydrocarbon concentration. This system is sensitive to changes in ambient parameters and therefore requires regular calibration. In this study, the FFID was calibrated at least twice a day to ensure the measurements accuracy. At all measurement locations, concentration data were collected for 180 seconds at 1000 Hz to achieve time convergence of the averaged concentrations. However, the effective experimental frequency response is dependent of the FFID capillaries used and was 78 Hz for background concentration measurements and 100 Hz for building sampling locations.

Ethylene mixed with nitrogen is used as a tracer gas and released in known quantities from the exhaust stack. The release rate is regulated by Bronkhorst mass-flow controllers to maintain a constant reduced-scale exhaust velocity of 7.4 m/s, representative of typical laboratory fume hood operating conditions. Pollutant emissions and ambient air are assumed to be isothermal; therefore, buoyancy effects are not

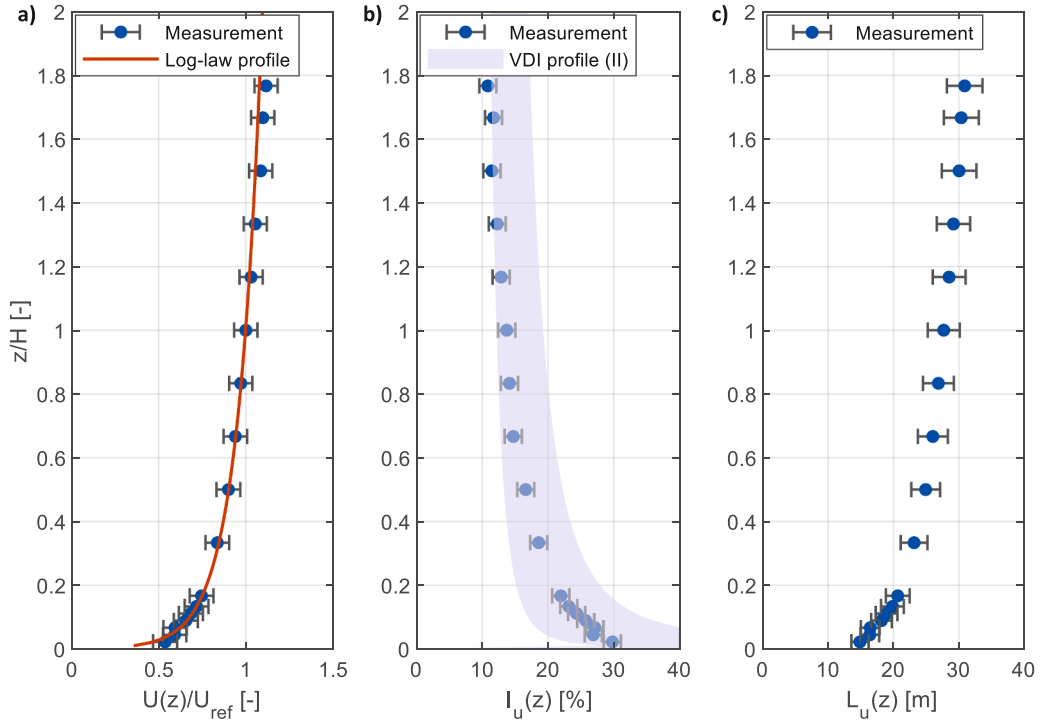


Fig. 4. Vertical profiles of (a) the normalized mean velocity $U(z)/U_{ref}$, (b) the longitudinal turbulence intensity $I_u(z)$ and (c) the integral turbulent length scale $L_u(z)$.

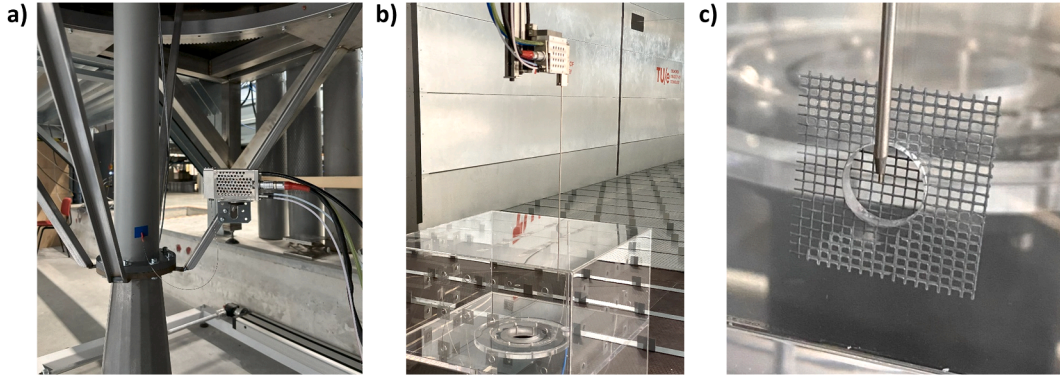


Fig. 5. Details of the tracer gas measurement positions a) in the exhaust fan duct, b) at an internal point, c) in front of a ventilation intake.

reproduced.

The accumulated pollutant concentrations re-introduced into the building interior through any air-inlets or leakages are measured at the exhaust pipe connecting to the fan as shown in Fig. 5a. Thirteen wind directions are considered: $\theta = 0^\circ, 15^\circ, 30^\circ, 45^\circ, 50^\circ, 60^\circ, 75^\circ, 90^\circ, 135^\circ, 180^\circ, 225^\circ, 270^\circ$, and 315° for the pollutant measurement at the exhaust fan pipe. The wind directions $\theta = 0^\circ$ and 45° were identified as the most critical (as will be discussed in Section 3.1); therefore, 27 additional measurement locations are added to obtain indoor pollutant concentration for these wind directions (Fig. 5b). For the most critical wind direction, $\theta = 45^\circ$, pollutant concentrations are also measured at each inlet (Fig. 5c). The internal measurement grid is a uniform $3 \times 3 \times 3$ grid in the reference frame defined in Figs. (3) and (6), with $x = [56.25; 225; 393.75]$ mm, $y = [56.25; 225; 393.75]$ mm, and $z = [-195; -120; -45]$ mm. The three cutting planes used to present the results are also shown in Fig. 6. Fig. 7a shows the measurement grid for the building internal measurement locations and Fig. 7b shows the location of the air-inlets. The exact coordinates of the internal points are provided in Appendix 1, and those of the 40 inlet measurement points are given in Appendix 2, defined in the same reference frame. This data can also be directly

downloaded in the dataset provided by Gillmeier and Jayakumari [49].

Tests are performed for three reference wind speeds, U_{ref} , defined as the mean wind speed at building height. Table 2 lists the three selected U_{ref} values along with their corresponding Reynolds numbers (Re). It has been verified that the flow remains turbulent after scaling, following the recommendations of Bahloul *et al.* [30]. These reference wind speeds are chosen to align with the publication of Jayakumari *et al.* [46] analysing the indoor-outdoor pressure characterization of the same building with the ventilation system. During pollutant measurements, U_{ref} is simultaneously monitored by measuring velocity recorded at the tunnel inlet with a 2-D ultrasonic anemometer and by applying a calibration factor. The resulting M values are also given in Table 2. The M value is defined as the ratio of exhaust speed (U_e) to reference wind speed (U_{ref}).

The results are analysed and presented in terms of normalized concentration, C^* , as defined in Eq. (2.3):

$$C^* = \frac{C - C_{background}}{C_{stack}} \quad (2.3)$$

where C is the measured pollutant concentration at any location (i.e., inside the exhaust pipe, inside the building or at the air-inlet) in ppm,

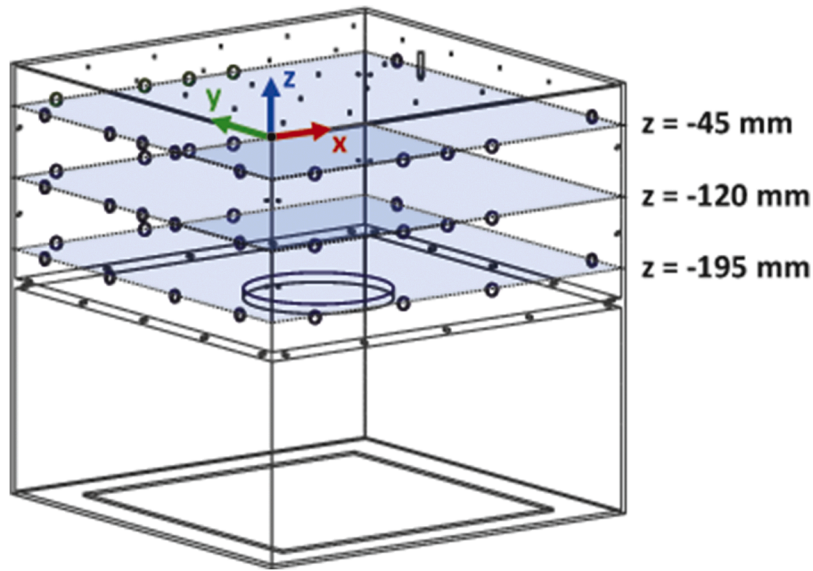


Fig. 6. Overview of the reference frame and the three cutting planes used to present the results.

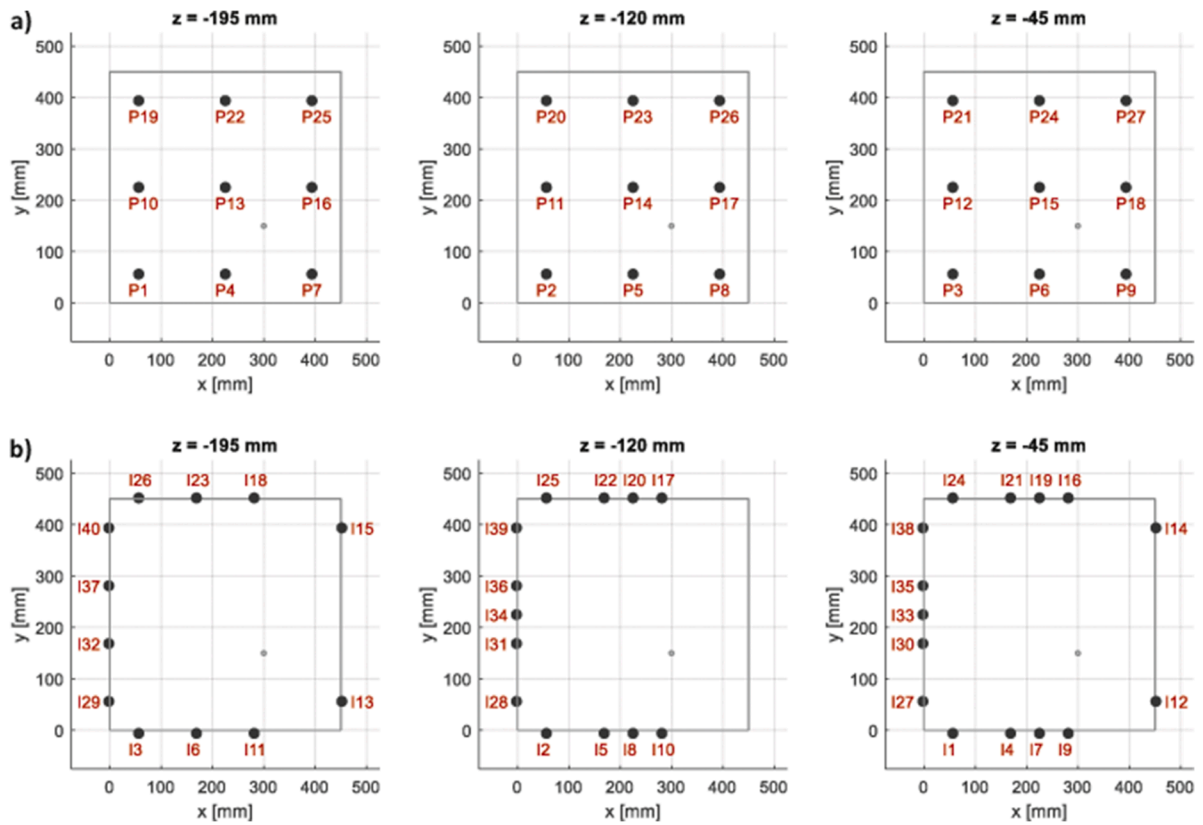


Fig. 7. Numbering of tracer gas measurement locations for a) the internal grid and b) the ventilation inlets.

$C_{background}$ is the background pollutant concentration measured in the approach flow in ppm, and C_{stack} is the known ethylene concentration in the released gas mixture through the stack. Three measurement repetitions are performed for several wind conditions (θ and U_{ref}), to quantify experimental measurement uncertainties. The repeatability error is thus computed for each wind condition as the largest difference between the replicates of C^* divided by the average value of C^* .

For the external mapping of the tracer-gas concentration, additional results are presented in terms of the intake-to-exhaust ratio R^* , defined

as the normalized concentration C^* at the ventilation intakes divided by the normalized concentration C^* measured in the exhaust pipe, as defined in Eq. (2.4):

$$R^* = \frac{C_{intake}^*}{C_{exhaust}^*} \quad (2.4)$$

where C_{intake}^* is the normalized concentration at the centre of a ventilation inlet, and $C_{exhaust}^*$ is the normalized concentration measured in the exhaust pipe.

Table 2

Reference wind speed and tracer gas release properties.

Parameter	Full scale	Reduced scale
Reference wind speed U_{ref}	4.5, 6.4, 7.8 m/s	6.4, 9.0, 11.1 m/s
Re based on L and U_{ref} ($> 11,000$)	5.4, 7.7, 9.4×10^6	1.9, 2.7, 3.3×10^5
Wind directions θ	0, 15, 30, 45, 50, 60, 75, 90, 135, 180, 225, 270, 315°	
U_e : Exhaust velocity	5.2 m/s	7.4 m/s
Released volume flow rate	1221 m ³ /h	1.08 m ³ /h
$M = U_e / U_{ref}$	1.2, 0.8, 0.7	1.2, 0.8, 0.7
Re based on D and U_e ($> 2,000$)	1.2×10^5	4.2×10^3

3. Results and discussion

3.1. Tracer gas concentration in the exhaust fan duct

To verify that the measured concentrations had reached time convergence, we first computed the cumulative average of the time series and checked that it remained, over a large portion of the record, within a tolerance band of ± 1 ppm around the final value, corresponding to the FFID system accuracy (see Section 3.4). As a representative example, Fig. 8 shows the cumulative average of the concentration difference $G - C_{background}$ measured in the exhaust fan duct as a function of time for $M = 1.2$ and the 45° wind direction. The tolerance band is also indicated by dotted lines in Fig. 8.

Normalized tracer gas concentrations are presented in Table 3 and illustrated as a spider plot in Fig. 9, as functions of wind direction and speed. The maximum relative deviation between the original measurement of C^* and its replicates is 12.7 %, obtained for the case at $\theta = 0^\circ$ and $M = 1.2$, and the average relative deviation is 5.4 % over the nine repeated measurements in the exhaust fan duct. The results indicate a strong dependency of pollutant re-introduction on wind conditions. Due to the intentional asymmetry of the ventilation layout, each wind direction represents a distinct aerodynamic configuration, yielding 13 different cases for model validation, each tested at three reference wind speeds. These experimental data highlight the variety of interactions likely to occur in real building settings between the tracer gas release, the wind flow and the ventilation system.

In particular, the M value has a pronounced effect on obtained results: for all tested wind directions, the proportion of reintroduced pollutant increases with higher M values, corresponding to lower reference wind speeds. For instance at $\theta = 45^\circ$, C^* decreases from 1.91×10^{-4} at $M = 1.2$ to 0.90×10^{-4} at $M = 0.7$, corresponding to a reduction of 53 % of the re-introduced concentration. However, this reduction coefficient is not the same for all wind directions, making it impossible to

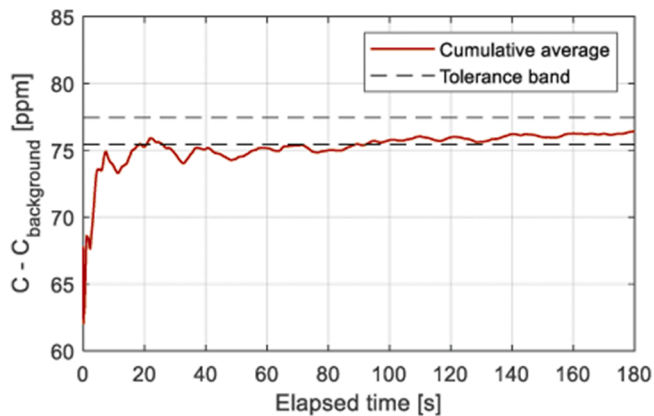


Fig. 8. Cumulative average of the concentration difference $C - C_{background}$ measured in the exhaust fan duct as a function of time for $M = 1.2$ and the 45° wind direction.

Table 3Re-introduced normalized tracer gas concentration C^* ($\times 10^{-4}$), measured in the ventilation exhaust duct.

	$M = 1.2$	$M = 0.8$	$M = 0.7$
0°	0.974	0.701	0.610
15°	1.679	1.018	0.722
30°	1.816	1.073	0.754
45°	1.911	1.267	0.898
50°	1.645	1.127	0.817
60°	1.161	0.864	0.660
75°	0.701	0.581	0.468
90°	0.322	0.230	0.190
135°	0.757	0.501	0.283
180°	0.184	0.133	0.093
225°	0.969	0.563	0.281
270°	0.508	0.329	0.254
315°	1.426	1.034	0.833

establish a clearer relationship between C^* and M values. This trend suggests that under weaker winds, the pollutant plume remains more coherent and is more easily recaptured by the facade air intakes, whereas stronger winds enhance dilution and plume deflection away from the building. Such behaviour is consistent with previous findings that re-introduction is favoured under low wind speeds and neutral buoyancy conditions [14].

As mentioned before, the wind direction also plays a crucial role: the reintroduced concentration varies by more than one order of magnitude between the most and least unfavourable wind direction (e.g. 1.91×10^{-4} at $\theta = 45^\circ$ vs. 0.18×10^{-4} at $\theta = 180^\circ$ for $M = 1.2$). The highest C^* values occur for oblique winds ($15^\circ < \theta < 50^\circ$), particularly $\theta = 45^\circ$, which aligns the exhaust plume with facades bearing the largest density of intakes, while building opposite winds ($\theta = 180^\circ$) result in lower interaction between the plume and the building.

A more detailed examination of the results highlights the strong dependence of plume behaviour and re-introduction on wind direction. For frontal winds ($\theta = 0^\circ$), the oncoming flow impinges directly on the windward facade, generating a stable wake that confines the tracer gas plume close to the upper part of the facade. While this promotes some interaction with the nearest ventilation intakes, lateral plume spreading remains limited, which explains the moderate values of re-introduced concentration. When the wind approaches obliquely ($15^\circ < \theta < 50^\circ$), the deflection of the plume along the roof becomes dominant: the flow separates at the upstream roof edge and reattaches asymmetrically, producing a large recirculation zone that also entrains the plume toward the facades containing the largest number of ventilation intakes. The peak re-introduction at $\theta = 45^\circ$ can therefore be attributed to the alignment between this dominant and wide recirculating structure and the layout of the ventilation openings. Moreover, the short distance between the tracer gas release and these openings also favors a lower dilution for this wind direction compared to others.

For perpendicular winds ($\theta = 90^\circ$ and $\theta = 270^\circ$), the asymmetric plume is clearly directed toward the facade with the lowest ventilation intakes density, explaining low re-introduced quantities. For opposite winds ($\theta = 180^\circ$), the exhaust plume is swept downstream, where it is rapidly diluted in the wake, with almost no recirculation toward the ventilation intakes. Winds from the diagonal directions ($\theta = 135^\circ$, 225° and 315°) show an intermediate behaviour depending on the number of facades exposed to the plume, the density of ventilation intakes in the upper part of these facades and the relative position of the exhaust stack compared to the downstream roof edges.

Overall, these observations confirm that re-introduction is driven by the interplay between building-induced flow recirculation and the positioning of ventilation openings relative to the plume dispersion. The strong contrasts observed between wind directions confirm that even modest changes in wind incidence can significantly modify the flow topology and, consequently, the risk of pollutant re-introduction.

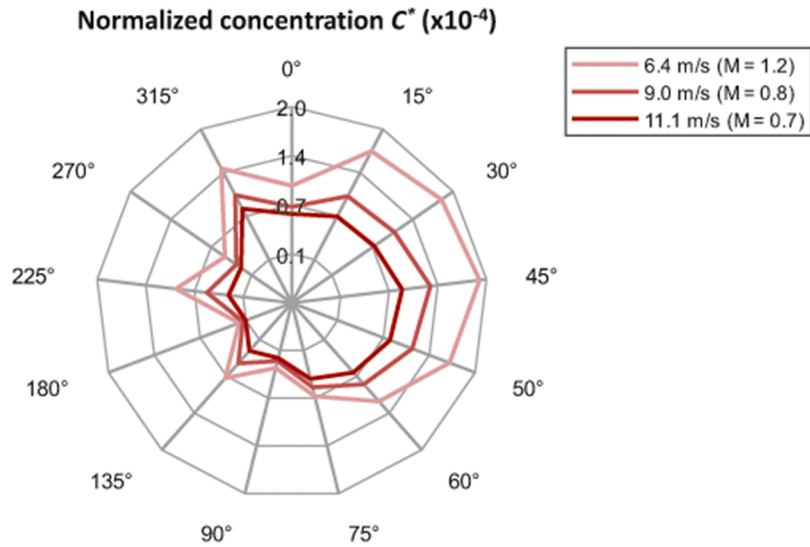


Fig. 9. Spider plot of the re-introduced normalized tracer gas concentration C^* ($\times 10^{-4}$), measured in the ventilation exhaust duct, as a function of wind conditions and speed.

3.2. Internal mapping of the tracer gas concentration

Internal concentration fields are shown in Fig. 10 and the corresponding data is supplied in Appendix 3. The maximum relative deviation between the original measurement of C^* and its replicates is 12.7 % for the internal mapping, obtained at P27 for the case at $\theta = 45^\circ$ and $M = 1.2$, and the average repeatability error is 9.0 % over the six repeated internal measurements. The results show that despite highly heterogeneous external concentrations, the tracer gas becomes rapidly well-mixed inside the building. This behaviour is consistent with the characteristics of the single-flow mechanical ventilation system, which continuously extracts air at a fixed rate while inducing compensatory inflows distributed over ventilation intakes. The combination of this

distributed outdoor air supply, the depressurization of the internal volume, and the small scale of the indoor space promotes intense mixing, leading to short internal residence times for large-scale concentration gradients. For $\theta = 0^\circ$, local internal time-averaged concentrations vary between 0.6 and 1.8 times the spatial and temporal averaged mean, while for $\theta = 45^\circ$ they range from 0.5 to 1.4, indicating lower spatial variability in the configuration resulting in the highest re-introduction rate. The observed variations indicate that no persistent concentration stratification or localized accumulation occurs, even under the most unfavourable wind conditions.

This homogenization is supported by the close agreement between the spatial and temporal averaged internal concentration and the temporal averaged exhaust duct concentration, with relative deviations of

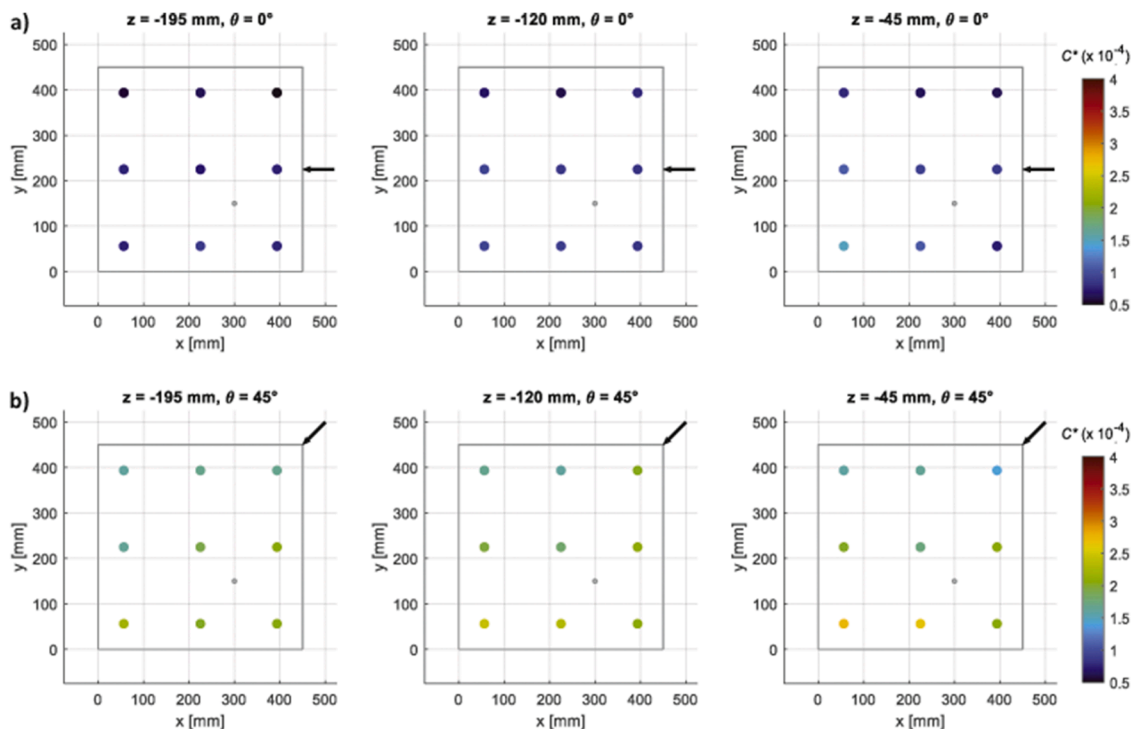


Fig. 10. Internal mapping of the normalized tracer gas concentration C^* ($\times 10^{-4}$) for $M = 1.2$ and a) $\theta = 0^\circ$, b) $\theta = 45^\circ$.

only 16 % ($\theta = 0^\circ$) and 3 % ($\theta = 45^\circ$). Such a correspondence suggests that, once the tracer enters the building through the ventilation inlets and leakages, it becomes well-mixed before reaching the extraction point. From a modelling perspective, this implies that an assumption of a uniform indoor contaminant field is reasonable for similar ventilation conditions, making it possible to use zonal or nodal approaches. This experimental data can also be used to assess CFD simulations of indoor flows. However, all mentioned modelling approaches require either a precursor simulation of the outdoor contaminant dispersion or boundary conditions for the contaminant at every ventilation opening. In this framework, the concentration of the tracer gas has been measured at each ventilation inlet as discussed in the next subsection.

3.3. External mapping of the tracer gas concentration

The tracer gas concentration at the centre of each ventilation inlet is provided in [Appendix 4](#) and displayed in [Fig. 11](#) for $\theta = 45^\circ$ and $M = 1.2$, representing the case with the highest re-introduction rate. The results confirm a strongly heterogeneous external distribution: inlets located downwind of the exhaust exhibit normalized concentrations above 3×10^{-4} , while those directly facing the wind show zero contamination. The external mapping also confirms the wide spreading of the tracer-gas plume observed at 45° for the lowest wind speed, highlighting the formation of a corner vortex and the enhanced recirculation developing along the leeward facades. The contrast between localized external concentrations and the uniform internal distribution underscores the critical role of the mechanical ventilation in controlling indoor dispersion. It also highlights a key challenge for numerical modelling: accurately resolving the interaction between a highly anisotropic external plume and a well-mixed indoor domain. The present dataset therefore provides a valuable benchmark for validating CFD, zonal and nodal models coupling outdoor dispersion with indoor ventilation networks simulations.

In this context, the intake-to-exhaust ratio R^* is reported in [Appendix 4](#). The ventilation intakes exposed to the tracer gas plume exhibit ratios between 1.449 and 2.040, whereas the other ones supply air with a zero tracer gas concentration. The average value of R^* is 1.063, meaning that the average concentration at the ventilation intakes differs by only 6.3 % from the concentration measured in the exhaust pipe. By definition, this ratio does not account for the 16 calibrated holes used to represent leakage, which likely explains the observed deviation combined with the measurement uncertainty. These findings confirm that the tracer gas concentration measured in the exhaust pipe reflects the integrated contribution of all inlet flows and leakage paths. The deviation between the average concentration at the ventilation inlets and the average indoor concentration is 3.4 %, which is consistent with the internal mapping analysis.

3.4. Limitations and future works

The present benchmark has been developed for a single, isolated and geometrically simplified building equipped with a single-flow mechanical ventilation system. As such, the conclusions are valid for this configuration only and do not explicitly account for the influence of surrounding buildings, street canyons or more complex urban canopy layouts. Addressing this relatively simple configuration was a necessary step before progressing towards more realistic cases, because it enables a clear interpretation of the wind-ventilation interaction without additional geometric complexity. Furthermore, following the methodology of Cook [52], the expected geometric scale for this case would be much smaller (e.g. 1:300–1:400); however, fully respecting the similarity requirements of the mechanical ventilation system imposed the choice of a 1:40 scale. This compromise implies that some large-scale turbulent structures of the atmospheric boundary layer cannot be perfectly reproduced in the wind tunnel and that the full-scale turbulence length scale is expected to be about four times larger than the one obtained experimentally, as discussed in previous work [47]. These scaling constraints introduce a limitation when extrapolating directly the results to other scales or to full-scale applications, even though the benchmark remains fully suitable to assess modelling approaches at the 1:40 scale.

In addition to geometric and aerodynamic constraints, several sources of experimental uncertainty must be acknowledged. Regarding the uncertainties associated with the wind-flow characterisation, the Cobra probe used in this study has a velocity measurement accuracy of ± 0.5 m/s, and the derived uncertainty in turbulence intensity is ± 1 %. The estimation of the turbulent length scale is more sensitive, as it depends on the threshold chosen for processing the velocity time series. The turbulence integral length scale is obtained by integrating the autocorrelation function of the velocity fluctuations, and varying the threshold between 0.01 and 0.05 leads to a relative deviation of 8.8%. The traverse system has a positioning accuracy of ± 1 mm. The impact of this uncertainty is expected to be negligible, as the concentration gradients over such small distances are limited. In any case, this contribution is implicitly accounted for within the repeatability uncertainty. The ventilation flow rate is determined from a differential pressure measurement, with a sensor accuracy of ± 2 Pa. Given the reduced-scale pressure magnitude around -40 Pa, this corresponds to a 5 % uncertainty in the pressure measurement, which translates into an uncertainty of 2.5 % in the exhaust airflow rate.

Another source of uncertainty concerns the FFID system. A constant gain of 500 was applied to both FFID channels throughout the measurements, resulting in a concentration range of approximately 0–3500 ppm. The accuracy of the voltage measurement is 2 mV, which corresponds to ± 0.7 ppm. At locations where zero concentration was expected, namely at ventilation intakes located upstream to the stack, the difference between the two FFID channels (background and current position concentrations) was ± 0.5 ppm in average, ranging from 0.3 to 0.6 ppm. This observation is consistent with the specified accuracy of

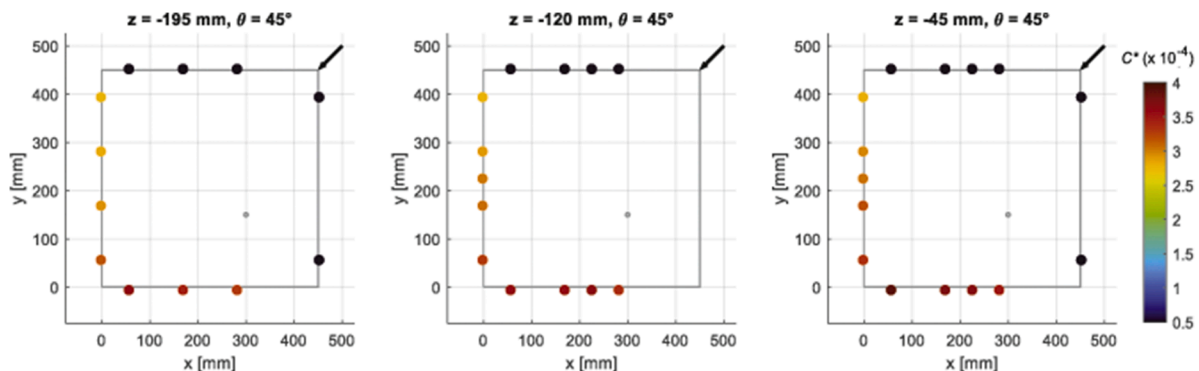


Fig. 11. External mapping of the normalized tracer gas concentration C^* ($\times 10^{-4}$) at the centre of ventilation inlets for $M = 1.2$ and $\theta = 45^\circ$.

± 0.7 ppm for each FFID channel. Because C^* is computed from the concentration difference between the two FFID channels, and assuming independent measurement errors, the uncertainty associated with C^* was obtained by combining the two contributions in quadrature, yielding an absolute uncertainty of ± 1 ppm. When expressed as relative deviations, this corresponds to an average of 2.8 % for the exhaust measurements, 6.3 % for the internal mapping at 0° (higher due to lower absolute concentrations), 2.6 % for the internal mapping at 45° , and 3.1 % for the external mapping at 45° .

The repeatability error is reported in the previous sections, with an average relative deviation of 5.4 % on C^* for the exhaust-pipe measurements and 9.0 % for the internal mappings. These values implicitly account for the combined effect of the uncertainty sources discussed above.

Future work should extend this experimental framework to more realistic and diversified scenarios, in order to broaden its applicability for both research and engineering practice. A first direction is to investigate more complex building geometries and layouts, including multiple volumes, varying dimensions, and configurations with surrounding structures representative of campuses or urban districts. This should be complemented by additional approach-flow conditions, such as different terrain roughness classes, non-neutral atmospheric stratification and non-isothermal releases, to quantify the combined impact of buoyancy and wind on pollutant re-introduction. On the modelling side, a dedicated parametric CFD study, once validated against the present benchmark, could be used to systematically vary key parameters such as stack height and location, exhaust velocity and M number, ventilation intakes layout and building airtightness. Such simulations would allow the individual contribution of each parameter to the re-introduction rate to be isolated, provide detailed visualisation of the associated flow patterns (e.g. corner vortices, roof recirculation, plume development), and support comprehensive sensitivity analyses aimed at identifying the most influential factors. Ultimately, combining extended wind-tunnel campaigns with parametric CFD studies will make it possible to build a robust experimental and numerical database for validating advanced pollutant dispersion models combining ABL flows with mechanical ventilation of indoor environments.

4. Conclusions

This study introduced a novel experimental benchmark to quantify the wind-driven re-introduction of pollutants into buildings through ventilation air intakes. By combining a reduced-scale building equipped with a controlled mechanical ventilation system, tracer gas source, and atmospheric wind conditions, the coupled dynamics of outdoor dispersion and indoor mechanical ventilation were reproduced experimentally. This unique setup enabled the systematic exploration of 39 combinations of wind directions and speeds, providing internal and external concentration fields under controlled conditions.

Results demonstrated that pollutant re-introduction is highly sensitive to wind conditions, with both direction and speed strongly influencing the amount of contaminant recaptured by the building. The most unfavourable configuration was found for oblique winds and low wind speeds, which maximized plume impingement on facades with numerous ventilation openings. Conversely, higher wind speeds promoted plume dilution and deflection away from the building, significantly reducing re-introduction rates. Despite the strong spatial heterogeneity of external concentrations, the contaminant became rapidly homogenized inside the building, highlighting the dominant mixing effect of the single-flow ventilation system. The consistency between mean indoor concentrations and those measured in the exhaust fan duct confirms this mixing behaviour.

Beyond its scientific contribution, this benchmark provides an original dataset for validating numerical models coupling outdoor dispersion and indoor ventilation using CFD, zonal, or nodal approaches. Such

validation is essential to ensure the reliability of simulations used for risk assessment and for the design of ventilation strategies minimizing pollutant re-introduction. The findings also have direct practical implications: they underscore the need to consider dominant wind directions and the positioning of ventilation openings, using accurate techniques during building design, especially in sites combining laboratory, industrial, and office spaces. Indeed, the ability to predict or quantify re-introduction rates is crucial for several scenarios of practical relevance. For instance, in laboratory buildings equipped with multiple fume hoods, this helps evaluate the risk of cross-contamination between adjacent rooms under varying wind conditions. In industrial facilities where exhaust stacks operate intermittently, this supports scenario-based assessments to determine whether certain operational modes (e.g., high-volume extraction under weak winds) may increase the re-introduction risk. Similarly, in densely built campuses, the quantification of pollutant re-introduction through ventilation openings offers a basis for assessing whether planned modifications—such as relocating air intakes, adding new exhaust stacks, or modifying stack dimensions—will reduce or exacerbate the risk of exposure to chemicals depending on the prevailing wind directions. Moreover, the present results allow engineering teams to test mitigation strategies numerically before implementation, once the numerical approach is validated against this experimental benchmark, such as optimizing stack discharge velocity, adjusting inlet shielding, or redesigning facade openings to avoid alignment with recirculation zones. All these possible applications demonstrate the potential of the re-introduction quantification to guide both design and retrofit decisions. More broadly, they support the importance of reducing emissions at their source, which remains the best way to prevent chronic exposure risks due to pollutant re-introduction and to preserve outdoor air quality.

A major limitation of the study is that results are expected to be specific for the building geometry (internal as well and external) and its design (including the mechanical ventilation characteristics), stack location and chosen approach flow characteristics. However, the asymmetrical design of the reduced-scale model still allows providing various combinations of wind conditions, pollutant source positions and ventilation openings positions. Future work should extend this experimental framework to more realistic configurations, including complex building geometries, surrounding structures, and non-isothermal releases. This stepwise approach aims to progressively build a robust knowledge base and provide the community with high-quality experimental data for validating pollutant dispersion and re-introduction predictions. Moreover, a parametric CFD study would indeed make it possible to isolate the contribution of each input parameter to the pollutant re-introduction rate and to visualise the associated flow patterns. Such a study would enable a sensitivity analysis aimed at identifying the factors that most strongly influence the re-introduction rate.

CRedit authorship contribution statement

Romain Guichard: Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Project administration, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Anjali Krishnan Radhakrishnan Jayakumari:** Writing – review & editing, Writing – original draft, Visualization, Validation, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Stefanie Gillmeier:** Writing – review & editing, Validation, Supervision, Resources, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization. **Ali Bahloul:** Writing – review & editing, Validation, Supervision, Methodology, Investigation.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence

the work reported in this paper.

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Appendix 1. Coordinates of the measurement locations used for internal mapping

Name	x [mm]	y [mm]	z [mm]
P1	56.25	56.25	-195
P2	56.25	56.25	-120
P3	56.25	56.25	-45
P4	225	56.25	-195
P5	225	56.25	-120
P6	225	56.25	-45
P7	393.75	56.25	-195
P8	393.75	56.25	-120
P9	393.75	56.25	-45
P10	56.25	225	-195
P11	56.25	225	-120
P12	56.25	225	-45
P13	225	225	-195
P14	225	225	-120
P15	225	225	-45
P16	393.75	225	-195
P17	393.75	225	-120
P18	393.75	225	-45
P19	56.25	393.75	-195
P20	56.25	393.75	-120
P21	56.25	393.75	-45
P22	225	393.75	-195
P23	225	393.75	-120
P24	225	393.75	-45
P25	393.75	393.75	-195
P26	393.75	393.75	-120
P27	393.75	393.75	-45

Appendix 2. Coordinates of the centres of ventilation inlets

Name	x [mm]	y [mm]	z [mm]
I1	56.25	-6	-45
I2	56.25	-6	-120
I3	56.25	-6	-195
I4	168.75	-6	-45
I5	168.75	-6	-120
I6	168.75	-6	-195
I7	225	-6	-45
I8	225	-6	-120
I9	281.25	-6	-45
I10	281.25	-6	-120
I11	281.25	-6	-195
I12	452	56.25	-45
I13	452	56.25	-195
I14	452	393.75	-45
I15	452	393.75	-195
I16	281.25	452	-45
I17	281.25	452	-120
I18	281.25	452	-195
I19	225	452	-45
I20	225	452	-120
I21	168.75	452	-45
I22	168.75	452	-120
I23	168.75	452	-195
I24	56.25	452	-45
I25	56.25	452	-120
I26	56.25	452	-195
I27	-2	56.25	-45
I28	-2	56.25	-120
I29	-2	56.25	-195
I30	-2	168.75	-45
I31	-2	168.75	-120

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Name	x [mm]	y [mm]	z [mm]
I32	-2	168.75	-195
I33	-2	225	-45
I34	-2	225	-120
I35	-2	281.25	-45
I36	-2	281.25	-120
I37	-2	281.25	-195
I38	-2	393.75	-45
I39	-2	393.75	-120
I40	-2	393.75	-195

Appendix 3. Internal normalized tracer gas concentration C^* ($\times 10^{-4}$) for $M = 1.2$ and wind directions of 0° and 45°

Name	0°	45°
P1	0.756	2.230
P2	0.948	2.459
P3	1.486	2.736
P4	0.881	2.011
P5	0.901	2.335
P6	1.074	2.651
P7	0.760	2.058
P8	0.851	2.094
P9	0.734	2.071
P10	0.772	1.631
P11	0.940	1.957
P12	1.092	2.015
P13	0.711	1.914
P14	0.865	1.808
P15	0.932	1.737
P16	0.754	2.068
P17	0.830	2.106
P18	0.889	2.078
P19	0.557	1.594
P20	0.683	1.666
P21	0.777	1.587
P22	0.648	1.680
P23	0.611	1.601
P24	0.654	1.627
P25	0.488	1.674
P26	0.785	2.019
P27	0.637	1.429

Appendix 4. Normalized tracer gas concentration C^* ($\times 10^{-4}$) and intake-to-exhaust ratio R^* at the centres of ventilation inlets for $M = 1.2$ and a 45° wind direction

Name	C^* ($\times 10^{-4}$)	R^*
I1	3.899	2.040
I2	3.563	1.864
I3	3.590	1.879
I4	3.722	1.948
I5	3.564	1.865
I6	3.431	1.795
I7	3.672	1.921
I8	3.639	1.904
I9	3.491	1.827
I10	3.373	1.765
I11	3.297	1.725
I12	0.000	0.000
I13	0.000	0.000
I14	0.000	0.000
I15	0.000	0.000
I16	0.000	0.000
I17	0.000	0.000
I18	0.000	0.000
I19	0.000	0.000
I20	0.000	0.000
I21	0.000	0.000
I22	0.000	0.000
I23	0.000	0.000

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(continued)

Name	C^* ($\times 10^{-4}$)	R^*
I24	0.000	0.000
I25	0.000	0.000
I26	0.000	0.000
I27	3.343	1.749
I28	3.279	1.716
I29	3.156	1.651
I30	3.193	1.671
I31	3.060	1.601
I32	2.931	1.534
I33	3.043	1.593
I34	3.028	1.585
I35	2.964	1.551
I36	2.893	1.514
I37	2.819	1.475
I38	2.792	1.461
I39	2.776	1.453
I40	2.769	1.449

Data availability

The data that support the findings of this study are openly available in *4TU.ResearchData* at <https://doi.org/10.4121/fb48f6a4-9e2c-491e-88b2-5aba8bbe9f06>

References

- [1] ACGIH, Industrial Ventilation: A Manual of Recommended Practice for Design, 31st Edition, 31st ed, American Conference of Governmental Industrial Hygienists, Cincinnati, Ohio, 2023. ISBN 978-1-60726-162-9.
- [2] ASHRAE, ASHRAE Handbook – Fundamentals, Chapter 24: Wind and Airflow Patterns Around Buildings, a, ASHRAE Editions, 2021.
- [3] ASHRAE, ASHRAE Handbook – HVAC Applications, Chapter 46: Building Air Intake and Exhaust Design, b, ASHRAE Editions, 2021.
- [4] H.D. Goodfellow, R. Kosonen (Eds.), Industrial Ventilation Design Guidebook (Second Edition) - Volume 1: Fundamentals. Industrial Ventilation Design Guidebook, Second Edition, Academic Press, 2020, pp. i–iii. ISBN 978-0-12-816780-9.
- [5] Industrial Ventilation Design Guidebook (Second Edition) - Volume 2: Engineering Design and Applications, in: H.D. Goodfellow, Y. Wang (Eds.), Industrial Ventilation Design Guidebook, Second Edition, Academic Press, 2021, pp. i–iii. ISBN 978-0-12-816673-4.
- [6] S. Dorregaray-Oyaregui, A. Ariño, L.A.D. Brito, et al., The conceptualization of exhalation in buildings, in: CLIMA 2022 conference, 2022.
- [7] A. Zuazua-Ros, L. de Brito Andrade, S. Dorregaray-Oyaregui, et al., Crosscutting of the pollutants and building ventilation systems: a literature review, *Env. Sci. Pollut. Res.* 30 (2023) 66538–66558, 66538–58.
- [8] S. Wu, Y. Wang, C. Chen, et al., Valley city ventilation under the calm and stable weather conditions: a review, *Build. Environ.* 194 (2021) 107668, 107668.
- [9] J. Quimbayo-Duarte, C. Chemel, C. Staquet, F. Troude, G. Arduini, Drivers of severe air pollution events in a deep valley during wintertime: A case study from the Arve river valley, France, *Atmos. Environ.* 247 (2021) 118030, 118030.
- [10] M. Chavez, B. Hajra, T. Stathopoulos, A. Bahloul, Near-field pollutant dispersion in the built environment by CFD and wind tunnel simulations, *J. Wind Eng. Ind. Aerodyn.* 99 (2011) 330–339, 330–9.
- [11] P.J. Drivas, P.G. Simmonds, F.H. Shair, Experimental characterization of ventilation systems in buildings, *Env. Sci. Technol.* 6 (1972) 609–614. American Chemical Society 609–14.
- [12] B. Hajra, T. Stathopoulos, A. Bahloul, The effect of upstream buildings on near-field pollutant dispersion in the built environment, *Atmos. Environ.* 45 (2011) 6724–6733, 6724–33.
- [13] B.K. Lamb, D.R. Cronn, Fume hood exhaust re-entry into a chemistry building, *Am. Ind. Hyg. Assoc. J.* 47 (1986) 115–123, 115–23.
- [14] D. Moon, A. Albergel, F. Jasmin, G. Thibaut, The use of the MERCURE CFD code to deal with an air pollution problem due to building wake effects, *J. Wind Eng. Ind. Aerodyn.* 67–68 (1997) 781–791, p. 781–91.
- [15] T. Stathopoulos, B. Hajra, M. Chavez, A. Bahloul, A Wind Tunnel Study of the Effect of Adjacent Buildings on Near-Field Pollutant Dispersion from Rooftop Emissions, IRSST, 2014.
- [16] C.J. van Leeuwen, Vermeire TG, Risk Assessment of Chemicals: An Introduction, Springer Science & Business Media, 2007. ISBN 978-1-4020-6101-1.
- [17] K. Ahmad, M. Khare, K.K. Chaudhry, Wind tunnel simulation studies on dispersion at urban street canyons and intersections—a review, *J. Wind Eng. Ind. Aerodyn.* 93 (2005) 697–717, 697–717.
- [18] M. Chavez, B. Hajra, T. Stathopoulos, A. Bahloul, Assessment of near-field pollutant dispersion: Effect of upstream buildings, *J. Wind Eng. Ind. Aerodyn.* 104–106 (2012) 509–515, 509–15.
- [19] D-P Guo, P. Zhao, W. Liu, Numerical and wind tunnel simulation studies of flow and pollutant diffusion around buildings under neutral and stable atmospheric stratifications, *J. Appl. Meteorol. Climatol.* 58 (2019) 2405–2420, 2405–20.
- [20] B. Hajra, T. Stathopoulos, A. Bahloul, A wind tunnel study of the effects of adjacent buildings on near-field pollutant dispersion from rooftop emissions in an urban environment, *J. Wind Eng. Ind. Aerodyn.* 119 (2013) 133–145, 133–45.
- [21] X.P. Liu, J.L. Niu, K.C.S. Kwok, J.H. Wang, B.Z. Li, Investigation of indoor air pollutant dispersion and cross-contamination around a typical high-rise residential building: Wind tunnel tests, *Build. Environ.* 45 (2010) 1769–1778, 1769–78.
- [22] J.H. Wang, J.L. Niu, X.P. Liu, C.W.F. Yu, Assessment of Pollutant Dispersion in the Re-entrance Space of a High-rise Residential Building, Using Wind Tunnel Simulations, *Indoor Built Environ.* 19 (2010) 638–647, 638–47.
- [23] J. Niu, T.C.W. Tung, On-site quantification of re-entry ratio of ventilation exhausts in multi-family residential buildings and implications, *Indoor Air* 18 (2008) 12–26, 12–26.
- [24] C.W. Tung, C.M. Mak, J.L. Niu, et al., Enlightenment of re-entry airflow: the path of the airflow and the airborne pollutants transmission in buildings, *Build. Environ.* 195 (2021) 107760, 107760.
- [25] T. Stathopoulos, L. Lazure, P. Saathoff, X. Wei, Dilution of exhaust from a rooftop stack on a cubical building in an urban environment, *Atmos. Environ.* 36 (2002) 4577–4591, 4577–91.
- [26] Ü Palmiste, J. Kurnitski, H. Voll, Design criteria for outdoor air intakes and exhaust air outlets located on an external wall, in: E3S Web Conf 172, EDP Sciences, 2020 09008, 09008.
- [27] US EPA, Residential Air Cleaners - A Technical Summary, US Environmental Protection Agency, 2018. EPA-402-F-09-002.
- [28] ASHRAE, ASHRAE Handbook – HVAC Applications, Chapter 44: Building Air Intake and Exhaust Design, ASHRAE Editions, 2007.
- [29] US EPA, User's Guide for the Industrial Source Complex (ISC3) Dispersion Model. Revised Volume II Description of Model Algorithms, US Environmental Protection Agency, 2004. EPA-454/b-95-0036.
- [30] A. Bahloul, T. Stathopoulos, M. Chavez, B. Hajra, IRSST Report; R-849, 2014.
- [31] B. Hajra, T. Stathopoulos, A. Bahloul, Assessment of pollutant dispersion from rooftop stacks: ASHRAE, ADMS and wind tunnel simulation, *Build. Environ.* 45 (2010) 2768–2777, 2768–77.
- [32] B. Hajra, T. Stathopoulos, A. Bahloul, Performance of ASHRAE models in assessing pollutant dispersion from rooftop emissions, *HVAC&R Res.* 20 (2014) 72–79, 72–9.
- [33] T. Stathopoulos, H. Bodhisatta, A. Bahloul, Analytical evaluation of dispersion of exhaust from rooftop stacks on buildings, *Rapp. Rech. Sci.* (2010).
- [34] B. Blocken, Y. Tominaga, T. Stathopoulos, CFD simulation of micro-scale pollutant dispersion in the built environment, *Build. Environ.* 64 (2013) 225–230, 225–30.
- [35] P. Gousseau, B. Blocken, G.J.F. van Heijst, CFD simulation of pollutant dispersion around isolated buildings: On the role of convective and turbulent mass fluxes in the prediction accuracy, *J. Hazard. Mater.* 194 (2011) 422–434, 422–34.
- [36] P. Gousseau, B. Blocken, T. Stathopoulos, G.J.F. van Heijst, CFD simulation of near-field pollutant dispersion on a high-resolution grid: A case study by LES and RANS for a building group in downtown Montreal, *Atmos. Environ.* 45 (2011) 428–438, p. 428–38.
- [37] X. Wang, K.F. McNamara, Evaluation of CFD Simulation using RANS turbulence models for building effects on pollutant dispersion, *Env. Fluid Mech.* 6 (2006) 181–202, 181–202.
- [38] J. Franke, A. Hellsten, K.H. Schlunzen, B. Carissimo, The COST 732 Best Practice Guideline for CFD simulation of flows in the urban environment: a summary, *IJEP* 44 (2011) 419. . 419.
- [39] R. Meroney, R. Ohba, B. Leitl, H. Kondo, D. Grawe, Y. Tominaga, Review of CFD guidelines for dispersion modeling, *Fluids* 1 (2016) 14, 14.

- [40] Y. Tominaga, T. Stathopoulos, CFD simulation of near-field pollutant dispersion in the urban environment: a review of current modeling techniques, *Atmos. Environ.* 79 (2013) 716–730, 716–30.
- [41] Y. Tominaga, T. Stathopoulos, Ten questions concerning modeling of near-field pollutant dispersion in the built environment, *Build. Environ.* 105 (2016) 390–402, 390–402.
- [42] B. Blocken, LES over RANS in building simulation for outdoor and indoor applications: a foregone conclusion? *Build. Simul.* 11 (2018) 821–870, 821–70.
- [43] T. Le Dez, J. Richard, C. Inard, et al., Reduced-scale study of the coupling between thermal and wind effects on the ventilation systems of nuclear facilities, *Int. J. Vent.* 20 (2021) 1–19, 1–19.
- [44] N. Le Roux, X. Faure, C. Inard, S. Soares, L. Ricciardi, Reduced-scale study of wind influence on mean airflows inside buildings equipped with ventilation systems, *Build. Environ.* 58 (2012) 231–244, 231–44.
- [45] N. Le Roux, X. Faure, C. Inard, S. Soares, L. Ricciardi, Reduced-scale study of transient flows inside mechanically ventilated buildings subjected to wind and internal overpressure effects, *Build. Environ.* 62 (2013) 18–32, 18–32.
- [46] A.K.R. Jayakumari, R. Guichard, S. Gillmeier, A. Ricci, B. Blocken, Atmospheric wind effects on depressurization for indoor asbestos pollution containment: Experimental analysis and a ventilation network model validation, *Build. Environ.* 272 (2025) 112664, 112664.
- [47] A.K.R. Jayakumari, S. Gillmeier, A. Ricci, R. Guichard, B. Blocken, Scaling effects on experimentally obtained pressures on an idealized building: possible implications towards asbestos containment, *J. Wind Eng. Ind. Aerodyn.* 239 (2023) 105442, 105442.
- [48] A.K.R. Jayakumari, R. Guichard, A. Ricci, S. Gillmeier, B. Blocken, Towards an improved wind effect assessment for asbestos abatement: a methodology for reduced-scale experiments, *J. Environ. Manag.* 370 (2024) 122698, 122698.
- [49] S. Gillmeier, A. Jayakumari, ERIES-Pollutant Reintroduction Investigation for Occupational Risk Reduction - a Benchmark. 4TU.ResearchData, Available at, <https://data.4tu.nl/datasets/fb48f6a4-9e2c-491e-88b2-5aba8bbe9f06>, 2025. Accessed 24 September 2025.
- [50] Environmental meteorology - Physical modelling of flow and dispersion processes in the atmospheric boundary layer - Application of wind tunnels, 2000.
- [51] HPAH. Irwin, The design of spires for wind simulation, *J. Wind Eng. Ind. Aerodyn.* 7 (1981) 361–366, 361–6.
- [52] N.J. Cook, Determination of the model scale factor in wind-tunnel simulations of the adiabatic atmospheric boundary layer, *J. Wind Eng. Ind. Aerodyn.* 2 (1978) 311–321, 311–21.